TCOP: An Agentic Early Warning and Response Protocol

Vishnu Vettrivel
Independent Researcher
vishnu@uofa.net

Abstract

An agent can consume untrusted content or obtain a credential in one domain and later request a consequential tool action in another. The observing domain may have useful signed evidence but no authority to enforce at the receiver. We present TCOP, an early warning and response protocol for AI agents, where the receiving domain decides what to do. TCOP carries signed, scoped, time-bounded runtime evidence to the receiver's local authorization boundary for a pending agent action. The receiver validates provenance, freshness, replay state, receipt correlation, subject, resource, capability, and scope before its own policy creates an authorization decision. TCOP specifies this evidence-to-authorization boundary, not a containment strategy. Under strict matched replay, 36 harmful actions forwarded without context changed to blocked with matched TCOP context, reducing harmful calls from 39 to 3. A single-environment reference path exercises the signed relay, receiver evaluator, and gateway as separate processes on one host. A mixed-action study shows that exact binding avoids the benign restrictions of a broad arrival guard but does not cover actions outside the accepted binding. In 100 policy-conditioned post-denial substitution episodes, Exact Binding permits all 15 eligible substitutions, whereas an evaluated receiver-local Campaign Correlation strategy blocks them and forwards the tested outside-campaign benign actions. A complete external-warning ledger separately reports detector coverage, receiver-actionable evidence, and harmful actions that proceed when no warning exists. These outcomes characterize configurable receiver strategies rather than mandatory TCOP behavior.

1 Introduction

Tool-using agents cross independently governed authorization boundaries. An agent can consume content or obtain a credential in one domain, then request a consequential tool or capability action in another. Prompt-injection research documents that untrusted content can steer tool-integrated agents toward direct harm or data exfiltration [19, 33]. The domain that first observes useful evidence may have no authority over the receiver's pending agent action.

Motivating pattern. Public accounts of a July 2026 incident involving OpenAI and Hugging Face illustrate the pattern: activity observed during one domain's evaluation later affected another organization's infrastructure, where the receiver detected and contained it [14, 25]. Those accounts do not establish whether origin evidence could have reached the receiver before its first harmful action, and this paper makes no incident-specific prevention claim.

The protocol question. The question is not whether an organization can publish or consume threat intelligence. It is whether signed evidence observed elsewhere should affect authorization of one pending agent action at a receiver. That requires a per-event contract: the receiver must check freshness, receipt correlation, subject, resource namespace, capability class, scope, and provenance against its own session and authorization state before its policy can use the evidence. TCOP, used here as a protocol name rather than an acronym, supplies this evidence-to-authorization contract and an action-relative timing model. It specifies evidence exchange and receiver acceptance, not the response strategy. Course of action, target selection, policy, and gateway enforcement remain with the receiving domain.

Figure 1 positions TCOP at the agent-action authorization boundary. An origin can send bounded runtime evidence to Domain B, but it cannot invoke Domain B's gateway or select a restriction. The receiver independently chooses a strategy, such as exact binding, monitoring, campaign correlation, or a broader guard, before its policy decides the pending action.

Timing determines whether imported evidence can affect a consequential agent action. A receiver can prospectively block an early action, reduce blast radius after an initial action, or retain late evidence for coordination after relevant actions have completed. This paper evaluates that timing dependence together with authority preservation, failure behavior, protocol validation, and availability cost.

The contributions are:

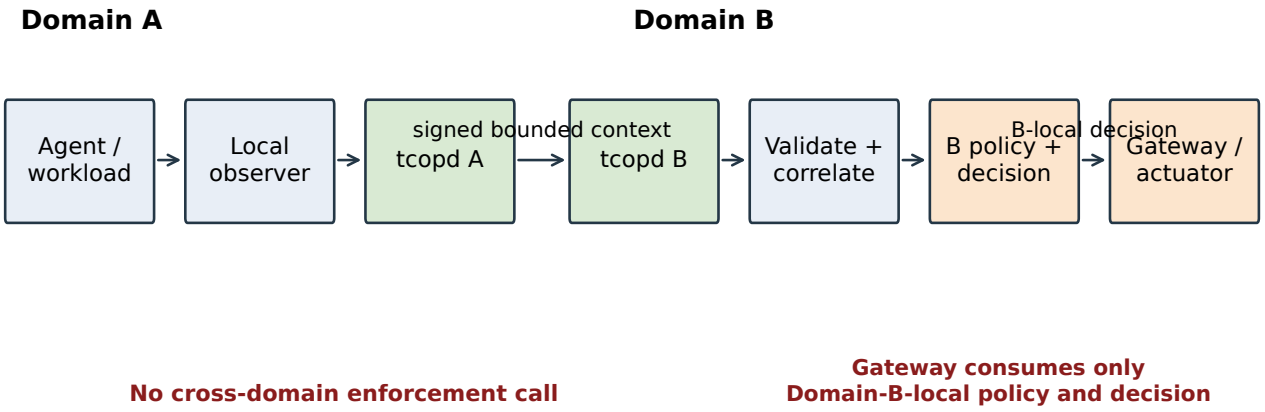


Figure 1: TCOP architecture. TCOP transports signed runtime evidence to a receiver-local authorization boundary; Domain A cannot invoke Domain B's gateway.

- a versioned runtime-evidence-to-authorization boundary for cross-domain agent actions, with receipt-bound, scoped, freshness-checked receiver acceptance, receiver-only decisions, and a separation between protocol acceptance and receiver strategy;
- a containment-window model and matched causal accounting that distinguish prospective blocking from subsequent blast-radius reduction; and
- an evidence package spanning deterministic pairs, frozen replay, a single-environment reference-path deployment with a gateway, a mixed-action validation cohort, post-denial substitution replay, external-warning coverage and admission, and executed receiver-validator mutations; hostile-peer and correlation tables are formula-computed illustrations.

2 Problem, Goals, and Threat Model

2.1 System model and design goal

We consider independently governed authorization boundaries that run agents or automation with local observers and enforceable local choke points. Domain A observes an event involving its own agent, workload, or telemetry. Domain B owns a downstream tool, resource, or gateway for a pending agent action. Domains may authenticate one another, but they do not share a policy engine or an enforcement API. Identity federation establishes who signed evidence; it does not confer authority to act in another domain.

TCOP permits runtime evidence observed in Domain A to influence a timely, scoped, and auditable Domain-B authorization decision for that pending action. The receiver can reject evidence, record a monitor-only disposition, restrict a narrow capability, or apply its own failure policy. The boundary follows the familiar policy-decision and policy-enforcement separation in zero-trust architectures while adding cross-domain runtime evidence [13, 28].

2.2 Adversary and trust assumptions

Agents, observers, and peer domains can be honest, compromised, or faulty. An adversary can relay, replay, delay, suppress, or reorder messages. An authorized peer can emit a syntactically valid but incorrect observation. A compromised client can present a valid receipt outside its intended session, and a stale local decision can survive a gateway restart unless the receiver checks current state. Domain B's enforcement boundary is trusted only to the extent that Domain B controls its policy and authorization adapter.

TCOP assumes a participating domain detects a useful fact, the receiver has an enforceable local agent-action choke point, and peer authorization plus public-key identity exist. The receiver binds accepted evidence to local state through a receipt or another locally governed mechanism. A signature authenticates a sender and protects the exchanged object; it does not establish observation truth or determine a response. This separation parallels remote-attestation architectures, which distinguish evidence from appraisal [4].

2.3 Evaluated properties and scope

The protocol goal is not to make a remote observation self-executing. TCOP evaluates five receiver-side protocol properties: authenticated provenance; freshness and replay resis-

Protocol property or evaluation rule	Receiver mechanism and scope
Authenticated provenance	Signature and authorized-issuer checks; mutation controls cover invalid signatures and unknown peers.
Freshness and replay resistance	Issued/expiry time and receiver replay state; expired and replayed controls create no restriction.
Correlation and scope integrity	Receiver-minted receipt plus subject, resource, capability, and scope comparison; concurrent-session and mismatch controls exercise the reference path.
Receiver sovereignty	Domain B creates the policy decision and gateway disposition; gateway blocks cite Domain-B-local decision records.
Failure ownership	Timeout, outage, monitoring, and restrictive behavior are receiver-policy outputs, not sender commands.
Evaluation accounting: non-retroactivity	Action-relative accounting never credits a late context with reversing a completed action; it is an outcome-accounting rule, not a cryptographic property.

Table 1: Protocol properties and evaluation rules. The final row bounds outcome accounting rather than protocol security.

tance; correlation and scope integrity; receiver sovereignty; and failure ownership. It evaluates these together with the non-retroactivity accounting rule. Table 1 maps each property and rule to the receiver mechanism and the evidence that supports its current claim boundary.

The tested receiver path checks these properties before an agent-action gateway disposition. Replayed, expired, unauthorized, or uncorrelated evidence was rejected in every tested control condition. The validation-value cohort adds executed subject and resource mismatch checks and action-like metadata mutations. Its hostile-peer and concurrent-session tables are formula-computed illustrations, not executed tests. It tests a reference implementation rather than proving the properties for all protocol implementations.

Table 17 concentrates the claim boundary. The paper evaluates synthetic tools and scenarios, receipt-based correlation, one reference gateway, local-harness performance, and a frozen model/runtime selection. It does not establish detection quality, observation truth, production-scale federation, global consensus, key-lifecycle operation, or rollback of completed actions.

3 Architecture and Protocol Boundary

Each domain runs a `tcopd` instance near its local observation and agent-action enforcement boundaries. Local telemetry reaches the local daemon; a signed `tcopd`-to-`tcopd` exchange carries TCX runtime evidence across domains; and the receiving daemon invokes a local authorization interface. Only the evidence exchange crosses domains. TCOP specifies that exchange and receiver acceptance; Domain B selects its strategy and policy. The gateway receives a Domain-B-local decision rather than the TCX object.

3.1 TCX context and receiver acceptance

We call the signed evidence record a TCX context: a versioned, scoped, time-bounded runtime-evidence object exchanged between domains. Figure 2 uses the canonical v0.2 record from the validation-value artifact. Before cross-domain delivery, Domain B registers an opaque receipt against its own bilateral interaction state. On receipt, the receiver parses and validates the record, checks freshness and replay state, resolves that receipt, and only then exposes a binding classification to local policy. The receipt is a Domain-B-minted correlation handle, not an authorization token.

The acceptance path makes the claimed mechanism reviewable. A valid signature alone is insufficient: Domain B checks the authorized issuer, version, issued and expiry time, replay state, receipt, and binding fields before evaluating its policy. The receiver compares subject, resource namespace, capability class, and scope with the pending agent action and its local authorization state. Exact binding is a local authorization input. A validated but non-exact object is not an input to the exact-binding strategy; a receiver may instead retain a monitor record or use a separately configured local strategy. Neither choice is prescribed by TCOP.

Exact Binding intentionally fails open for actions outside an accepted binding. If a detector emits no warning or the receiver cannot establish an applicable binding, Exact Binding has no TCOP-derived basis to restrict that action. A receiver can combine it with a separate local guard, but the guard's availability cost is a receiver-policy choice and is reported separately.

Campaign Correlation (artifact identifier C2E): Campaign Correlation is one receiver-local strategy, not a TCOP requirement. Invalid, expired, or replayed context is rejected before a restriction. Valid but unrelated or sender-suggested-only campaign information creates no restriction or only a monitor record. Only receiver-established session lineage or a receiver-recorded delegation edge can activate Campaign Correlation; the sender neither supplies the relation nor selects its scope or disposition. The frontier exercises five created, matched, and de-escalated relations plus one expiry. These controls characterize the declared strategy, not a protocol-mandated lifetime.

3.2 Authority separation and lifecycle

After accepting evidence, Domain B combines it with locally held facts, selects a locally configured strategy, evaluates a local policy, and creates a local decision record for the pending agent action. The gateway consumes that decision through a local authorization-evaluator interface. Imported fields remain evidence fields; Domain B selects the response scope and failure disposition.

The evaluated cryptographic profile uses Ed25519 signatures and protocol version v0.2. Peer authorization and the

Illustrative redacted TCX v0.2 artifact example

```
{ protocol_version: "0.2",
issuer: "origin-monitor.example",
observation_id: "demo-observation-42",
receipt_reference: "sha256:demo-receipt",
subject: "agent-demo-17",
resource_namespace: "demo-target",
capability_class: "repository.write",
scope: "guarded-capability",
issued_at: "2030-06-01T12:00:00Z",
expires_at: "2030-06-01T12:01:00Z",
signature: "ed25519:example" }
```

Receiver acceptance and decision

```
for c in contexts(receipt):
if !valid(c): reject(c); continue
if exact_binding(c, action): exact += c
else: partial += c
return local_policy(action, local_state, exact, partial)
```

Branches: invalid or unresolved evidence → rejection; exact binding → a local authorization input; partial binding → a receiver-local monitor or escalation choice. No sender field selects the disposition.

Receiver audit state. `receipt registered` → `received` → `parsed` → `signature checked` → `issuer authorized` → `freshness checked` → `replay checked` → `receipt resolved` → `binding matched` → `B-local policy` → `B-local decision` → `gateway disposition`. Any failed transition emits a rejection audit record and creates no restriction. A repeated verified context is rejected by the receiver replay check before receipt resolution. For Campaign Correlation only, receiver-created relation state follows `created` → `matched` → (`deescalated` or `expired`); sender-suggested relations are monitor-only.

Figure 2: Illustrative redacted TCX artifact example and receiver acceptance and decision path. The receiver validates an object before classifying its binding to the pending action. Partial evidence is not a remote command: a receiver-local strategy and policy may ignore, monitor, or respond to it.

Table 2: TCOP design requirements.

Requirement	Mechanism
Signed provenance	Signed TCX context with authorized peer identity
Bounded scope and expiry	Scoped, time-bounded context validation
Replay protection	Receiver replay checks and fresh local decisions
Receiver-local correlation and policy	Domain B correlates and applies its own strategy and policy
No remote enforcement command	Imported context never maps directly to allow, deny, quarantine, revoke, or suspend
Capability-scoped response	Domain B restricts only locally selected capabilities
Audit linkage	Receipt and decision records link context to local action
Partition-local operation	Local controls remain available during federation failure
Explicit availability cost	Benign restriction impact is measured, not hidden

effective verification keys are receiver-local configuration. Key rotation, revocation dissemination, and inter-domain key-distribution operation are specified deployment concerns but are outside the evaluated evidence. The paper therefore treats the protocol as a concrete reference boundary, not a complete federation lifecycle standard.

3.3 Partition behavior and disclosure

Federation failure suppresses new imported evidence. Domain B retains its local observer, policy, and enforcement path, and its local policy determines the failure disposition. The design exports enough evidence to verify provenance, scope, freshness, and correlation without exporting the origin's full monitor state, policy, or a gateway command. Attribute disclosure, retention, peer authorization, and response duration remain federation-policy choices.

4 Receiver-Local Resolution and Enforcement

The receiving pipeline is runtime-evidence validation, local correlation, receiver-local strategy selection, policy evaluation, decision creation, and gateway enforcement for a pending agent action. The ordering matters. Evidence that cannot be correlated to a current local subject cannot affect an unrelated session, and accepted evidence has no direct mapping to an enforcement action.

4.1 Local decision path

The receiver evaluates the chain: accepted imported evidence → local correlation → policy evaluation → local decision → gateway authorization. The implementation records the receipt reference, accepted evidence, policy identity, decision identity, and enforcement result. This makes a changed agent-action authorization traceable to the receiver's own decision path rather than to the arrival of a remote message alone.

The gateway adapter calls a local authorization-evaluator interface and does not embed TCX resolver or strategy logic. This mirrors the policy-decision and policy-enforcement split in zero-trust architecture [28]. The reference enforcement point is a Docker MCP Gateway tool-call boundary; MCP's own authorization flow remains responsible for client access [21].

4.2 Validation and failure handling

The validation-value study exercises receiver checks through 11 deterministic mutations. Signature, issuer, version, receipt, expiry, and replay faults created no restriction in every case exercised. A valid message that contained action-like `action` or `enforcement` metadata was accepted only as evidence; those remote fields were ignored when the receiver created its local decision. Table 7 reports this and distinguishes the

Table 3: Complementary system contracts. TCOP supplies a receiver-side evidence-to-authorization boundary; it does not replace threat intelligence, response orchestration, tool access, or agent collaboration protocols.

System class	Cross-domain object	Bound to pending action?	Receiver validates local action state?	Sender selects response?
STIX/TAXII	Indicators, reports, collections	No	No	No
CACAO/SOAR	Playbooks and courses of action	No	Deployment-specific	May carry a requested response
MCP	Tool invocation and authorization context	At tool boundary	Server access context	Client requests tool
A2A	Tasks, messages, and artifacts	Task-related	Not a security-evidence contract	Agent requests task
TCOP	Signed runtime evidence	Yes	Yes	No

formula-computed correlation and hostile-peer illustrations from executed evidence.

Authorization caching is disabled for correctness and counterfactual replay. Each authorization checks current receiver-local state, preventing an old decision from being reused after a context change. A separate fixed-time-to-live cache selection measures local overhead without changing the correctness path.

The receiver also owns failure behavior. If `tcopd-B` is unavailable or the authorization interface times out, the sender does not choose fail-open or fail-closed semantics. Domain B applies the disposition in its local policy. The design requires fresh state after gateway restart, but restart remains untested. The executed stale-decision control checks recomputation.

4.3 Authority evidence

The 30 single-environment reference-path arms exercise signed federation, validation, evaluation, and gateway composition. Every gateway block cites Domain-B-local policy and decision provenance. This is not an adversarial remote-enforcement test; the executed action-metadata mutation supplies that misuse control.

5 Containment Windows and Outcome Accounting

A warning changes an outcome only if it becomes actionable before the receiver-local authorization boundary of an affected agent action. We call this interval a *containment window*. It is action-relative: a warning can arrive too late for one completed operation and still affect a later agent action in the same workflow.

Figure 3 shows the three accounting regimes. Before the first affected authorization, imported runtime evidence can support prospective blocking. After an initial harmful agent action but before later opportunities, it can reduce subsequent blast radius. After all relevant opportunities complete, it can remain useful for forensic or coordination work but is not credited as prevention.

The evaluation therefore does not label an evidence receipt, policy change, or alert as a success by itself. A causal improvement requires accepted runtime evidence, a bound

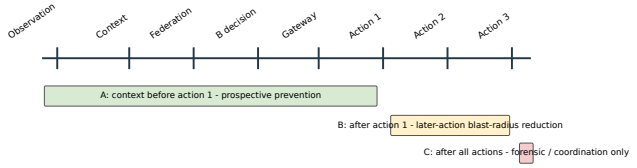


Figure 3: Action-relative containment-window regimes. A warning can be too late for one action and still be actionable for a later one.

local subject, a local policy evaluation, a local decision, and a changed gateway disposition for the same agent-action opportunity. Late evidence is recorded separately from changed outcomes.

This timing model serves two purposes. It prevents a late warning from being counted as a reversal of completed harm, and it exposes the timing precondition that any cross-domain early-warning system must satisfy. The single-environment reference-path deployment and replay studies vary delivery placement around these boundaries while holding the underlying synthetic action sequence fixed.

6 Matched Causal and Conformance Evaluation

6.1 Evaluation questions and evidence hierarchy

The evaluation tests action-time authorization, receiver authority, timing, and binding selectivity. Deterministic matched pairs supply the primary causal comparison; credentialed execution generates frozen traces for strict delivery-timing replay. Single-environment reference-path arms test federation and gateway composition. The sealed-plan validation-value study executes binding and receiver-validation checks; its hostile-peer and correlation tables are formula-computed, not measured. Separately rooted studies test post-denial branches and external warnings. Table 4 defines their accounting units.

Table 4: Units of analysis and accounting. Counts are not interchangeable across rows.

Unit	Definition	Use in this paper
Strict pair	One input-equivalent A1/A2 comparison with the same frozen local policy and action schedule	Primary deterministic causal comparison
Trace	One captured model-generated action sequence	Live cohort and frozen replay input
Replay treatment	One frozen trace evaluated at one action-relative context-delivery placement	Timing and availability accounting
Single-environment reference-path arm	One complete source-relay-evaluator-gateway execution in one reference environment	Gateway composition and authority provenance
Gateway call	One authorization request for an action opportunity	Harmful and benign forwarded or blocked outcomes
Signed exchange	One signed context transfer from `tcopd`-A to `tcopd`-B	Relay composition accounting
Validation episode	One credential-free mixed-action test case per condition	C0–C3 validation-value cohort

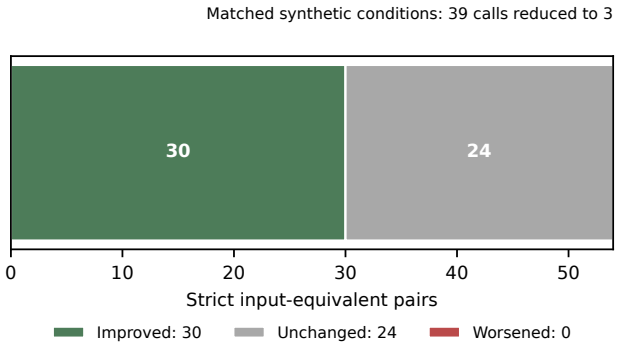


Figure 4: Deterministic strict-pair outcomes. The full matched population includes improved and unchanged pairs. The zero worsened count follows from the containment-first monotonicity of the reported harmful-action measure.

6.2 Deterministic matched design

The deterministic evaluation compares isolated capable local observers (A1) with the same receiver consuming accepted imported evidence (A2), fixing local facts, policy, strategy, action inputs, and schedule. Each pair records input equivalence, warning margin, local decision change, harmful-action difference, and outcome direction. This synthetic matrix tests causal conformance, not operational prevention.

For containment-first, the study includes 54 strict input-equivalent pairs. Imported context changes 30 outcomes and leaves 24 unchanged. Under strict matched replay, 36 harmful actions forwarded without context changed to blocked with matched TCOP context, reducing harmful calls from 39 to 3. Because containment-first can only add a restriction to the reported harmful-action measure, 0 worsened pairs is a monotonicity check, not a surprising comparative result. Figure 4 shows the unchanged population rather than presenting only improved pairs.

The three residual harmful calls are the three strict resilience rows with no imported context or later harmful opportunity recorded. They are unchanged rather than late-warning failures, so the study does not attribute them to TCOP.

Table 5: Strict deterministic containment-first comparison. Balanced and utility configurations with unmatched local policies are excluded.

Metric	Value
Strict input-equivalent pairs	54
Improved / unchanged / worsened	30/ 24/ 0
Baseline / treatment harmful calls	39/ 3
Harmful actions changed to blocked under matched replay	36/39
Unmatched-policy exclusions	123

6.3 Admission rules and central control

The study excludes 123 candidate balanced and utility comparisons from A1/A2 causal inference because their receiver-local policies differ. The exclusion ledger retains their inputs and the mismatch reason. This admission rule protects the question of whether imported context changed an outcome under the same local policy.

A fact-equivalent central audit holds exportable facts fixed while placing decision authority or an outage dependency in a central path. Across 54 audited cells, it records 6 regressions: 3 associated with authority placement and 3 with central outage; 48 cells show no regression. It identifies dependencies of this construction, rather than serving as a general benchmark of centralized defense.

6.4 Validation value beyond arrival

The validation-value study holds receiver enforcement, action order, local facts, and warning arrival fixed across Local Only (C0), Arrival Default Allow (C1-allow), Arrival Guard (C1-class), Exact Binding (C2), and Standing Guard (C3). The study follows a sealed pre-analysis plan. The arrival conditions receive a valid-arrival token and a time-to-live. Exact Binding additionally evaluates issuer, signature, version, freshness, receipt, subject, resource namespace, capability class, scope, and receiver-local authorization state.

Table 6 reports the entire 12-episode mixed-action cohort. Arrival Default Allow forwards every action, showing that

Table 6: Mixed-action validation-value cohort governed by a sealed pre-analysis plan. Full-cohort harmful counts include mismatched-warning controls; the accepted binding-matched subset is reported separately in the text.

Condition	Harmful blocked	Benign constrained
Local Only (C0)	0/12	0/12
Arrival Default Allow (C1-allow)	0/12	0/12
Arrival Guard (C1-class)	7/12	7/12
Exact Binding (C2)	3/12	0/12
Standing Guard (C3)	12/12	12/12

Table 7: Validation-value evidence: executed receiver mutations and formula-computed illustrations.

Check	Result
Valid hostile-peer burst	Formula-computed illustration; no executed burst evidence.
Receipt correlation ambiguity	Formula-computed illustration with typed-in zeros; no measured correlation result.
Receiver-validator mutations	11 mutations cover signature, issuer, version, receipt, expiry, and replay faults; action-like remote fields are retained as evidence and ignored for enforcement.

arrival by itself is not a containment policy. Standing Guard establishes the standing-restriction endpoint. Its complete harmful blocking comes with complete benign constraint.

Across the full cohort, Arrival Guard blocks 7 of 12 harmful actions and constrains 7 of 12 benign actions. Exact Binding blocks 3 of 12 harmful actions and constrains 0 of 12 benign actions. Within the accepted, binding-matched harmful subset, Arrival Guard and Exact Binding each block 3 of 3 matching harmful actions. The additional Arrival Guard blocks occur under wrong-subject, wrong-resource, wrong-receipt, expired, replayed, and semantic-smuggling warning controls, which Exact Binding rejects. This is the measured selectivity result, not evidence that Exact Binding is a broader class guard.

6.5 Correlation under ambiguity

Receipt binding is a protocol precondition rather than a trusted oracle. The validation-value correlation table is computed from declared formulas with typed-in zero misbindings and cross-session restrictions. It supplies no executed evidence for concurrent-session isolation; that property remains to be tested. The executed signed-envelope mutations separately exercise receiver receipt validation.

6.6 Result interpretation

An improved pair records a changed harmful-action outcome. Unchanged pairs include completed actions, non-actionable evidence, and equal baseline outcomes. Availability is evaluated separately. This design establishes receiver-path causality and conformance, not post-denial behavior prevalence or deployment effectiveness.

7 Agent, Replay, and Gateway Evaluation

7.1 Live trace generation and admission

A credentialed model generated 44 eligible traces in bounded synthetic prompt-injection, credential/tool-misuse, and benign metadata-write scenarios, with no recorded model error or refusal. Eligibility depends on captured action opportunities, not refusal or safe early termination. Freezing the traces separates generation from enforcement causality and removes model nondeterminism from paired replay.

7.2 Strict replay and timing

Strict replay fixes captured actions, local policy, strategy, gateway configuration, and receiver monitors, varying only accepted evidence availability and timing. It records receipt, correlation, decision change, gateway outcome, and the affected action opportunity.

The strict replay matrix contains 220 paired comparisons. Figure 5 reports action-relative placements. Early and boundary placements represent the tested preauthorization region; later placements can affect only later opportunities. The complete treatment table and normalized timing summary retain their denominators in Appendix E.

7.3 Post-denial substitution replay

The separately rooted post-denial substitution study compares receiver-local strategies through the same TCOP acceptance boundary. It holds 100 policy-conditioned replay episodes fixed across 20 frozen traces per policy. Every imported-evidence policy blocks the initial exact-bound action; 15 traces then have an eligible harmful capability, session, or delegation substitution. Exact Binding (C2) permits all 15 substitutions, whereas Arrival Guard (C1), Campaign Correlation (C2E), and Standing Guard (C3) block them. Campaign Correlation uses a predeclared receiver-local relation, not a sender-provided enforcement field. Each imported-evidence strategy forwards all 5 of 5 exercised predeclared safe fallback actions. These safe fallbacks are a distinct in-campaign benign-continuation branch that the frozen policies explicitly forward after the initial denial. They are not the same-risk outside-campaign benign population on which the frontier measures Arrival Guard and Standing Guard collateral restriction.

The study also retains 12 bounded reference-runtime episodes. The runtime follows a predeclared harmful substitution branch after denial in 9 episodes and a safe fallback in 3 episodes. These runtime observations establish branch representability in the evaluated environment. The strict replay supplies the policy-conditioned conformance result; neither component estimates behavior prevalence in deployment.

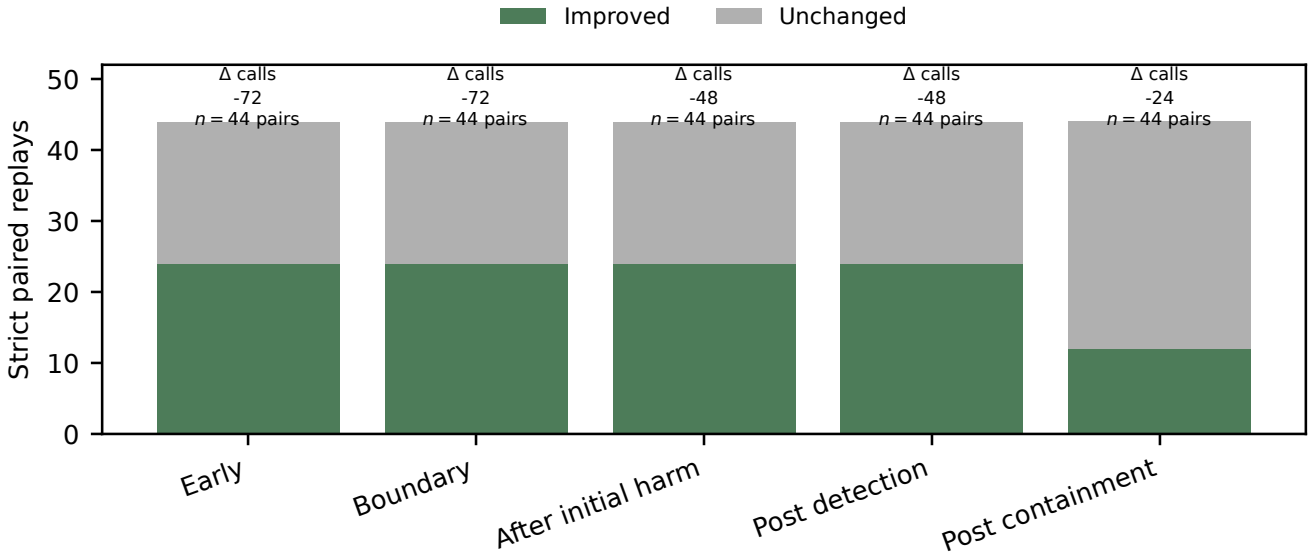


Figure 5: Strict replay results by action-relative delivery placement. Early and boundary treatments share the tested discrete preauthorization region.

Table 8: Receiver-local strategy frontier. The same 10 harmful and 20 benign logical actions are evaluated per policy; this is a deterministic reference-policy result.

Strategy	Harmful blocked	Benign constrained
Local Only (C0)	0/10	0/20
Arrival Guard (C1)	10/10	10/20
Exact Binding (C2)	5/10	0/20
Campaign Correlation (C2E)	10/10	0/20
Standing Guard (C3)	10/10	10/20

Campaign Correlation strategy frontier: The separately sealed frontier evaluates 150 deterministic policy episodes over 10 harmful and 20 benign logical actions. Its composition is fixed before policy execution (Table 9): exact-bound and non-exact harmful actions share the `sensitive-write` risk family; the latter carry a receiver-established session-lineage relation. The same-risk benign actions outside a campaign retain that risk family but have a receiver-local campaign miss. Arrival Guard and Standing Guard block the same harmful population but constrain 10 and 10 benign actions; Exact Binding blocks only 5 harmful actions. Campaign Correlation blocks all 5 campaign-linked non-exact harmful substitutions and forwards all 5 benign same-risk actions outside the campaign. The receiver policy contract permits a declared delegation edge, but this frontier exercises session lineage only. Invalid, expired, and replayed context is rejected. Valid sender-suggested-only context is monitor-only, and the study records 0 unrelated restrictions.

Table 9: Campaign Correlation frontier input composition (artifact identifier C2E). The fixed population contains five traces in each principal scenario; session lineage is the only strong relation exercised in this frontier.

Fixed trace class	Harmful	Benign
Exact-bound action	5	0
Non-exact action with receiver session lineage	5	0
Predeclared safe fallback after relation	0	5
Same-risk action outside campaign	0	5
Unrelated subject, session, or resource	0	5
Invalid relation	0	2
Expired relation	0	1
Replayed relation	0	1
Sender-suggested-only relation	0	1

7.4 Single-environment reference-path deployment

The single-environment reference path runs `tcopd`-A, signed exchange, `tcopd`-B, the evaluator, and MCP gateway as separate processes on one host. It evaluates neither multi-host nor administrative separation. The 20 signed exchanges traverse both endpoints. Figure 7 separates prospective blocking from completed calls and later blocked opportunities. Each arm is one complete execution; harmful call attempts and signed exchanges are separate accounting units. Every gateway block cites Domain-B-local policy and decision provenance, establishing composition of this reference path.

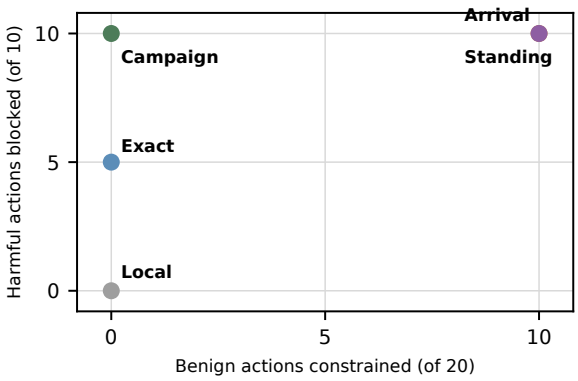


Figure 6: Receiver-local policy frontier in the sealed deterministic Campaign Correlation study. Arrival Guard and Standing Guard overlap at 10 benign actions constrained and 10 harmful actions blocked. The plotted points characterize configurable receiver strategies, not a mandatory TCOP response.

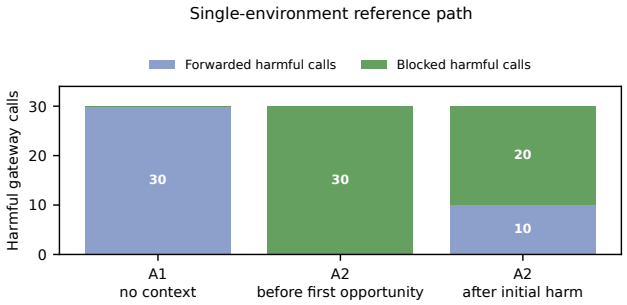


Figure 7: Single-environment reference-path timing result. The late arm separates already forwarded harmful calls from subsequent calls blocked by a Domain-B-local decision.

7.5 Measured scope

The reported sub-millisecond values characterize the local deterministic harness. The single-environment reference-path deployment establishes that the tested relay, evaluator, and gateway adapter compose; strict replay supplies the controlled counterfactual. The timing treatments vary delivery relative to a receiver-local authorization boundary in synthetic action sequences. They do not reconstruct or draw measurements from a public incident.

8 Security and Availability

8.1 Failure controls and receiver validation

Ten live controls were executed and passed. Unknown peer and remote TCX action were not injected live; executed signed-envelope mutations cover them separately. Disallowed import type, no remote enforcement, gateway restart, and session termination before arrival remain untested design properties despite recorded passes. Appendix E preserves every row and its evidence classification. Exercised malformed or stale contexts create no restriction; late context receives no prevention credit, and timeout or outage resolves through Domain B's policy.

Table 10: Single-environment reference-path gateway arms. Each block is attributable to a Domain-B-local decision.

Arm	Forwarded	Blocked
A1, no context	30	0
A2, before first opportunity	0	30
A2, after initial harm	10	20

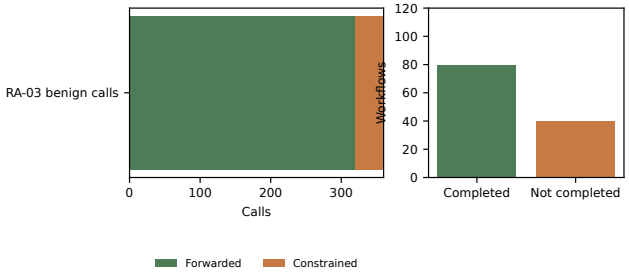


Figure 8: Measured availability cost in the selected RA-03 synthetic workload.

8.2 Valid hostile peers

Signature validation authenticates the issuer; it does not establish that the issuer's observation is true. The sealed hostile-peer burst table is computed from declared formulas, not from executed warning bursts. Its idempotence and quota-plus-monitor entries illustrate candidate receiver policies but establish no measured overload behavior. Hostile-peer behavior remains to be tested.

8.3 Security-availability trade-off

Containment-first has a material availability cost in the selected synthetic false-warning treatments. In the inside-window early and boundary treatments, 40 of 120 benign agent calls are constrained and 0 of 40 affected workflows complete. The remaining no-evidence and late treatments forward 240 of 240 benign agent calls and complete 80 of 80 workflows. The aggregate 80 of 120 result is reported for accounting, but does not blend away the treatment-specific false-warning cost.

The reported cohort establishes a bounded receiver-strategy frontier. Never restricting forwards the selected benign workload. Standing Guard maximizes blocking but constrains the capability continuously. Arrival Guard activates the same

Table 11: Measured RA-03 availability cost under selected synthetic false-warning treatments and capability restriction; this is not a universal false-positive rate.

Treatment group	Attempted	Forwarded	Constrained	Workflow completion	Affected capability
Inside-window early and boundary	120	80	40	0/40	repository.write in frozen benign metadata workflow
No context, outside window, post-detection, and post-containment	240	240	0	80/80	no restriction in these treatments
All selected treatments	360	320	40	80/120(66.7%)	selected synthetic capability restriction

class-wide guard after accepted context becomes actionable. Exact Binding preserves containment on the accepted exact-binding subset while narrowing the restriction to that binding. The post-denial substitution and frontier studies add Campaign Correlation, which covers the declared related substitutions that Exact Binding alone permits.

8.4 External-warning coverage and admission

A credential-free successor evaluates the complete sealed external ledger from the pinned AgentDojo commit [9] and pinned Prompt Guard 2 86M output configuration [20]; it consumes preserved detector outputs rather than rerunning the detector. Its declared normalizer maps `LABEL_0` to no warning and `LABEL_1` to a TCX prompt-attack category. Table 12 separates source coverage, detector output, normalization, receiver admission, and exact binding. Of 72 attack-bearing source cases, 25 produce a warning and an admitted exact binding; 47 produce no warning and therefore no actionable TCOP evidence. TCOP cannot compensate for a detector no-warning result.

Table 13 reports every policy over all 168 external source cases. Arrival Guard, Exact Binding, and Campaign Correlation block 25 of 72 harmful actions, the 25 for which accepted exact evidence exists, and forward the remaining 47 no-warning harmful actions. They forward all 96 benign no-warning actions. Standing Guard blocks 72 of 72 harmful actions but constrains all 96 of 96 benign actions because it is a standing receiver guard. This is an integration and authorization-coverage result, not an estimate of detector quality, warning prevalence, or deployment effectiveness.

Table 14 adds deliberately derived protocol controls to the selected external bases. Exact Binding and Campaign Correlation apply no restriction to partial or mismatched bindings. Stale and replayed context are rejected before policy. Sender-suggested-only relation information remains nonrestrictive under Campaign Correlation. Campaign Correlation forwards all 12 harmful substitutions without a receiver-created relation, including 8 session-lineage and 4 delegation variants; it blocks all 12 paired substitutions when the receiver has created the corresponding local relation. These controls demonstrate the declared receiver logic, not a general claim about correlation quality.

Table 12: Full external-warning pipeline coverage. All 168 cases in the sealed detector-output ledger are evaluated; a warning is not a detector-quality estimate.

Stage or source category	Cases
Attack-bearing source cases	72
Detector warning → normalized TCX → admitted exact binding	25/25/25
Detector no-warning → no actionable TCOP evidence	47
Benign source cases, all detector no-warning	96
Total no actionable evidence	143

8.5 Local overhead

The performance experiment reports authorization, gateway, and warning-to-enforcement measurements separately for cache-disabled and fixed-time-to-live cache selections. The correctness and causal replays use the cache-disabled configuration. The local-harness measurements appear in Appendix E.

9 Related Work

9.1 Threat-intelligence exchange

STIX/TAXII, MISP, and DShield distribute threat information [10, 23, 24, 29]. TCOP may ingest evidence derived from them; it addresses the later agent-runtime question of whether a receiver should authorize one pending tool or capability action after receipt-, session-, and action-specific validation. These systems do not specify that action-time receiver-local authorization contract.

9.2 Response orchestration

RID, DOTS, and OpenC2 exchange incident reports, mitigation signals, and defense commands [22, 26, 27]. CACAO and SOAR coordinate playbooks and response workflows [6,16]. Either can implement a response after TCX acceptance. TCOP instead bounds the preceding decision input: whether signed runtime evidence applies to one pending receiver-side agent action. Course of action, workflow execution, and enforcement remain receiver-local.

Table 13: Full-population receiver authorization outcomes over 72 attack-bearing and 96 benign external source cases. Arrival Guard, Exact Binding, and Campaign Correlation use accepted evidence conditionally; Standing Guard is a standing local restriction.

Policy	Harmful blocked	Harmful forwarded	Benign constrained	Benign forwarded
Local Only (C0)	0/72	72/72	0/96	96/96
Arrival Guard (C1)	25/72	47/72	0/96	96/96
Exact Binding (C2)	25/72	47/72	0/96	96/96
Campaign Correlation (C2E)	25/72	47/72	0/96	96/96
Standing Guard (C3)	72/72	0/72	96/96	0/96

Table 14: Independent Warning Admission v2 derived controls. Each row contains 12 derived cases. The substitution rows contain 8 session-lineage and 4 delegation variants.

Control	Receiver result
Partial binding	Exact Binding/Campaign Correlation: 0/12 restrictions
Mismatched binding	Exact Binding/Campaign Correlation: 0/12 restrictions
Stale or replayed context	Campaign Correlation: 12/12 rejected before policy
Sender-suggested-only relation	Campaign Correlation: 0/12 restrictions
Harmful substitution, no B-local relation	Campaign Correlation: 12/12 forwarded
Harmful substitution, B-local relation	Campaign Correlation: 12/12 blocked

9.3 Trust, identity, and local policy

KeyNote and macaroons support delegated authorization [3, 5]; TCX instead carries behavioral runtime evidence into a locally created decision. This avoids imported evidence becoming ambient enforcement authority, related to the confused-deputy concern [12]. Zero trust separates policy decision from enforcement [28]; SPIFFE federation authenticates foreign workloads [31]; and RATS separates evidence, appraisal, verifier, and relying party [4]. TCOP applies those separations to cross-domain agent actions: it combines authenticated peer identity with receiver-local correlation and appraisal. Collaborative intrusion detection identifies relevant coordination and adversarial-participant questions [7]; the central audit here is only a bounded fact-equivalent control.

9.4 Agent interoperability and tool protocols

MCP standardizes client access to server tools and resources [21]; its gateway is the reference tool-call enforcement point here. A2A standardizes independent-agent discovery and collaboration [1], and ANP addresses agent-network communication and discovery [2]. They can supply transport or session context. Table 3 summarizes the contract-level distinction: TCOP adds receipt-bound validation of signed runtime evidence against a pending receiver-side agent action, while leaving response selection and enforcement with the receiver.

9.5 Shared Signals and agent authorization

SSF, CAEP, and RISC carry identity, credential, and session state changes between parties as Security Event Tokens [8, 15, 30, 32]. AIMS gives best practices for agent identity, credentials, and OAuth-based authorization; its Section 11 permits Shared Signals subscriptions and recommends recipient remediation, including access attenuation or renewed policy evaluation [17]. AAP covers identity assertions, scoped grants, delegation, short-lived behavioral attestation presented with credentials, and cross-organization credential revocation pushed to federation members [11]. The Liu framework composes OAuth 2.0 extensions for delegated agent authorization at token issuance, with audit records for post-hoc review [18]. These mechanisms carry identity and session state or decide authorization at grant or token time; AIMS and AAP also include runtime signals. TCOP carries what another domain observed an agent doing, bound to a specific pending action at the receiver, where it can only tighten authority.

9.6 Agent and tool security

Prompt injection and tool-integrated agents motivate the synthetic scenarios. Liu et al. benchmark prompt-injection attacks and defenses, while InjecAgent evaluates indirect injection against tool-integrated agents [19, 33]. TCOP does not compete with those detectors. It evaluates whether externally observed runtime evidence can affect a locally governed pending action in time.

10 Limitations and Conclusion

10.1 Limitations

The claim boundary in Section 2 and Table 17 applies throughout. Evidence is limited to synthetic tools and scenarios, receipt-based correlation, one reference gateway, local-harness performance, and one frozen model/runtime selection. Post-denial branches and Campaign Correlation selectivity are bounded reference-strategy results. External-warning results consume preserved detector outputs, not estimates of detector quality, warning prevalence, or production effectiveness. Exact Binding's equal containment holds only on the accepted binding-matched subset, not the full cohort. Signed-envelope

mutations establish only the exercised receiver behavior; hostile-peer and correlation tables are formula-computed, with typed-in zero correlation counters, not measured evidence. Key rotation, revocation propagation, heterogeneous federation, and cross-host transport delay remain unevaluated. The visible RA-03 eligibility correction (Table 16) affects benign availability; the separately rooted strict causal comparison excludes RA-03.

Future work: runtime bindings across hosts. TCOP defines abstract tightening actions once and binds them per runtime, separating native and harness-built primitives without pooling results. The study compares a bare agent (E0), the same agent in a sandbox runtime (E1), and TCOP bound to it (E2). Deny-by-default egress already blocks unknown destinations; for an allowed but misused destination, only receiver-bound evidence can act, and only if its binding takes effect before the action. A poll-delayed binding prevents nothing for earlier actions; the harness instant cut tests that gap. A transport binding could carry a TCX as a Security Event Token over Shared Signals [15, 32]. See the sealed repository amendments `spec/external-warning-adaptive-crosshost-amendment-003.md` through `spec/external-warning-adaptive-crosshost-amendment-005.md`.

10.2 Conclusion

TCOP makes timely cross-domain evidence an input to receiver-local authorization. The evaluated strategies expose a precision, coverage, and availability frontier rather than a mandatory protocol response. In every tested single-environment reference-path and replay condition, gateway blocks cite receiver-local policy and decision records. The artifact preserves source roots, exclusions, sealed plans, and regenerated results so each claim can be checked against its evidence.

References

[1] A2A Project. Agent2agent (a2a) protocol specification. A2A Protocol v1.0, 2026. Version 1.0.1 current at access; accessed August 2, 2026.

[2] Agent Network Protocol. Agent network protocol. Protocol overview, 2026. Overview page; accessed August 2, 2026.

[3] Arnar Birgisson, Joe Gibbs Politz, Úlfar Erlingsson, Ankur Taly, Michael Vrable, and Mark Lentczner. Macaroons: Cookies with contextual caveats for decentralized authorization in the cloud. In *Network and Distributed System Security Symposium*. Internet Society, 2014.

[4] Henk Birkholz, Dave Thaler, Michael Richardson, Ned Smith, and Wei Pan. Remote ATtestation procedureS (RATS) architecture. RFC 9334, Internet Engineering Task Force, 2023.

[5] Matt Blaze, Joan Feigenbaum, John Ioannidis, and Angelos D. Keromytis. The KeyNote trust-management system version 2. RFC 2704, Internet Engineering Task Force, 1999.

[6] Robert A. Bridges, Ashley E. Rice, Timothy S. Oesch, Jeffrey A. Nichols, Cory L. Watson, Kevin D. Spakes, Savannah A. Norem, Michael R. Huettel, Brian C. Jewell, Brian Weber, Connor M. Gannon, Olivia M. Bizovi, Samuel C. Hollifield, and Samantha H. Erwin. Testing SOAR tools in use. *Computers & Security*, 129:103201, 2023.

[7] Rainer Bye, Seyit Ahmet Camtepe, and Sahin Albayrak. Collaborative intrusion detection framework: Characteristics, adversarial opportunities and countermeasures. In *Workshop on Collaborative Methods for Security and Privacy*. USENIX Association, 2010.

[8] T. Cappalli and A. Tulshibagwale. OpenID Continuous Access Evaluation Profile 1.0. OpenID Foundation, https://openid.net/specs/openid-caep-1_0-final.html, August 2025.

[9] Edoardo Debenedetti, Jie Zhang, Mislav Balunović, Luca Beurer-Kellner, Marc Fischer, and Florian Tramèr. Agentdojo: A dynamic environment to evaluate prompt injection attacks and defenses for LLM agents. In *Advances in Neural Information Processing Systems*, 2024.

[10] Alexandre Dulaunoy and Andras Iklody. MISP core format. Internet-Draft, draft-20, November 2025. Work in progress; accessed August 2, 2026.

[11] A. Fane. OpenA2A Agent Authorization Protocol (AAP). Internet-Draft draft-fane-opena2a-aap-02, https://datatracker.ietf.org/doc/draft-fane-opena2a-aap/02/, September 2026. Individual submission, work in progress.

[12] Norman Hardy. The confused deputy: Or why capabilities might have been invented. *ACM SIGOPS Operating Systems Review*, 22(4):36–38, 1988.

[13] Vincent C. Hu, David F. Ferraiolo, Rick Kuhn, Adam Schnitzer, Kenneth Sandlin, Robert Miller, and Karen Scarfone. Guide to attribute based access control (ABAC) definition and considerations. Special Publication 800-162, National Institute of Standards and Technology, 2014.

[14] Hugging Face. Anatomy of a frontier lab agent intrusion: A technical timeline of the july 2026 incident. Hugging Face technical timeline, July 27, 2026. Accessed August 2, 2026.

[15] P. Hunt, M. Jones, W. Denniss, and M. Ansari. Security Event Token (SET). RFC 8417, https://www.rfc-editor.org/rfc/rfc8417, July 2018.

[16] Bret Jordan, Allan Thomson, Vasileios Mavroeidis, and Mateusz Zych. CACAO security playbooks version 2.0. OASIS Committee Specification 01, 2023.

[17] P. Kasselman, J. Lombardo, Y. Rosomakho, B. Campbell, N. Steele, and A. Parecki. AI Identity Management System. Internet-Draft draft-ietf-wimse-aims-00, IETF WIMSE Working Group, https://datatracker.ietf.org/doc/draft-ietf-wimse-aims/00/, September 2026. Work in progress.

[18] D. Liu, H. Zhu, S. Krishnan, A. Parecki, and H. Xue. AI Agent Authorization Integration Framework. Internet-Draft draft-liu-ai-agent-authorization-integration-00, https://datatracker.ietf.org/doc/draft-liu-ai-agent-authorization-integration/00/, July 2026. Individual submission, work in progress.

[19] Yupei Liu, Yuqi Jia, Runpeng Geng, Jinyuan Jia, and Neil Zhenqiang Gong. Formalizing and benchmarking prompt injection attacks and defenses. In *33rd USENIX Security Symposium*. USENIX Association, 2024.

[20] Meta Llama. Llama prompt guard 2 86m model card. Hugging Face model card, 2025.

[21] Model Context Protocol. Authorization. Model Context Protocol Specification, 2025-06-18, 2025. Versioned specification URL; accessed August 2, 2026.

[22] Kathleen Moriarty. Real-time inter-network defense (RID). RFC 6545, Internet Engineering Task Force, 2012.

[23] OASIS Cyber Threat Intelligence Technical Committee. STIX version 2.1. OASIS Standard, 2021.

[24] OASIS Cyber Threat Intelligence Technical Committee. TAXII version 2.1. OASIS Standard, 2021.

[25] OpenAI. OpenAI and Hugging Face partner to address security incident during model evaluation. OpenAI public incident update, July 21, 2026. Accessed August 2, 2026.

[26] Tirumaleswar Reddy, Dan Wing, Mohamed Boucadair, and Jan Medved. Distributed denial-of-service open threat signaling (DOTS) signal channel specification. RFC 9132, Internet Engineering Task Force, 2021.

[27] Jason Romano and Duncan Sparrell. Open command and control (OpenC2) language specification version 1.0. OASIS Committee Specification Draft, 2018.

[28] Scott Rose, Oliver Borchert, Stu Mitchell, and Sean Connelly. Zero trust architecture. Special Publication 800-207, National Institute of Standards and Technology, 2020.

[29] SANS Internet Storm Center. Using DShield's data feeds. SANS Internet Storm Center documentation, 2026. Accessed August 2, 2026.

[30] M. Scurtescu, A. Backman, P. Hunt, J. Bradley, S. Bounev, and A. Tulshibagwale. OpenID RISC Profile Specification 1.0. OpenID Foundation, https://openid.net/specs/openid-risc-1_0-final.html, August 2025.

[31] SPIFFE Community. SPIFFE federation. SPIFFE Standard, 2026. Versioned specification URL; accessed August 2, 2026.

[32] A. Tulshibagwale, T. Cappalli, M. Scurtescu, A. Backman, J. Bradley, and S. Miel. OpenID Shared Signals Framework Specification 1.0. OpenID Foundation, https://openid.net/specs/openid-sharedsignals-framework-1_0-final.html, August 2025.

[33] Qiusi Zhan, Zhixiang Liang, Zifan Ying, and Daniel Kang. InjecAgent: Benchmarking indirect prompt injections in tool-integrated large language model agents. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 10471–10506. Association for Computational Linguistics, 2024.

A Open Science Appendix

The source and artifacts are available at https://github.com/cloudronin/tcop. It contains the paper source, source-verification reports, normalized data, figure and table sources, expected results, claim and number audits, a compact reviewer package, the sealed TCX Validation-Value Study v2, Post-Denial Substitution Study v1, Campaign Correlation Frontier Study v1 (artifact identifier C2E), Independent Warning Admission v2 external-stratified study, and the full-population external-warning successor. The predecessor roots remain unchanged for lineage. It contains no provider credentials, production credentials, personal paths, or unpublished incident data.

The artifact is organized around four reproduction tiers. Tier 0 verifies immutable roots, normalizes data, regenerates figures and tables, audits manuscript numbers and anonymity, and builds the paper. Tier 1 reruns the deterministic causal core without external credentials. Tier 2 strictly replays frozen agent traces through local-only and TCOP treatments without model credentials. Tier 3 is an optional credentialed regeneration of live traces and is not required for the causal findings.

The default reviewer command runs Tier 0 plus strict agent replay without reading a model-provider environment variable. The validation-value study is credential-free and includes its sealed output, pre-analysis plan, expected results, mutation corpus, and two byte-identical clean rerun report. The post-denial substitution, Campaign Correlation, and independent-warning studies include sealed plans, normalized summaries, source manifests, candidate and selection ledgers where applicable, and dedicated verifier commands documented in the anonymous artifact. The expected-results manifest records the strict-pair counts, single-environment reference-path arms, control outcomes, strategy digests, source roots, validation-value results, and all study results used by the manuscript.

The artifact uses Python, the TCOP CLI, Matplotlib for vector figures, and a TeX distribution for the PDF. Docker is required only for the optional reference-gateway smoke path, which also requires a separately supplied pinned gateway checkout. The source-verification report records the formal amendment that admits the available deterministic evidence root in place of an unavailable predecessor root.

Tier 0, Tier 1, Tier 2, and the validation-value verifier complete without provider credentials. Tier 3 documents its provider/runtime lock, eligibility criteria, bounded attempts, expected variability, and cost category separately from frozen-trace reproduction. A fresh provider run is not permitted to replace the replay evidence used for the paper's causal claims.

B Ethics Appendix

All evaluated tools and target systems are synthetic. No production credentials, real repositories, customer systems, datasets, or workloads were attacked. Prompts and tool calls are restricted to a test environment, and no autonomous production enforcement occurred. The credentialed model activity was bounded by synthetic tool interfaces. No human subjects were involved, and the study does not collect or release personal data.

The work concerns defensive handling of potentially harmful tool calls. An executable gateway and replay environment could be misused if presented as a way to control another domain. The artifact reduces this risk through synthetic inputs, omitted target infrastructure and credentials, and a receiver-local authorization boundary. The evaluated protocol does not export a remote enforcement interface or a production attack path.

False or malicious warnings are the principal ethical and operational risk. The paper reports the measured benign-workflow disruption caused by containment-first restriction. The formula-computed hostile-peer burst table supplies no measured evidence of overload behavior. A receiver decides whether to restrict, monitor, deduplicate, or rate-limit a warning. This cost and responsibility remain part of the design result.

The public incident is a motivation for the cross-domain failure pattern, not an attribution or empirical reconstruction.

C Frozen Strategy Details

The study certifies four objective-specific frozen configurations: containment-first, balanced federated, utility-preserving, and forensic-oriented. The paper's primary causal deterministic comparison uses containment-first because it is the configuration with matched local-policy pairs. Balanced and utility configurations are retained in the artifact but their unmatched comparisons are excluded from the causal claim. Artifact configuration identifiers remain in the machine-readable strategy manifest.

The full manifest digests and certification status appear in the source-verification report and normalized strategy records. The paper does not portray the forensic configuration as a separable composable overlay, and it makes no unsupported reconstruction-quality claim.

D Live Amendment History

The live evaluation preserves a visible sequence of sealed artifacts. The earliest provider-failure artifacts remain negative results. Amendments correct provider transport compatibility, align frozen benign eligibility, correct an artifact completion gate, reconcile a derived utility counter from immutable replay rows, and add the reference-path origin relay without new model sampling. The immutable historical artifact identifier retains its original end-to-end label. None changes TCX fields, receiver-local policies, scenario prompts, tools, treatment definitions, or frozen traces.

D.1 RA-03 eligibility correction

Amendment 003 corrected an eligibility implementation that applied a global high-risk capability rule to the benign RA-03 metadata-write workflow. The frozen benchmark had already declared `repository.write` permitted for that workflow. The original and corrected rules, as well as their distinct retained trace-eligibility reports, are included in the artifact. Table 16 gives the sensitivity view required to distinguish a trace-eligibility correction from a changed harmful-action result.

E Extended Results

The normalized data preserves the full strict-pair and treatment-level rows with source artifact digests, paths, and row identifiers. The timing result distinguishes delivery before the first harmful opportunity, at the first-action authorization boundary, after initial harm but before later actions, after receiver-local detection, and after partial local containment.

Table 15: Live runtime amendment history. Digest prefixes identify the immutable predecessor and successor roots; full digests are retained in the normalized amendment-chain data and source-verification report. The first two predecessor artifacts are retained provider-failure negatives.

Amendment	Cause	Model sampling?	Traces?	Protocol / policy?	Result derivation?	Predecessor digest	Successor digest
001	Provider completion-token field	No	No successful sampling	No	No	e5cf49806b70	698ca3c116af
002	Provider reasoning mode	No	No	No	No	698ca3c116af	f9bccfcaeff8
003	RA-03 eligibility aligned to frozen truth	Yes, eligible cohort capture	No	No	Eligibility only	f9bccfcaeff8	cebc669a7a3c
004	Completion-gate ordering	No	No	No	No	cebc669a7a3c	24a20e9c48a2
005	Derived utility reconciliation	No	No	No	Yes, replay-row aggregation	24a20e9c48a2	2b1f92ff4481
006	End-to-end tcopd-a relay replay	No	No	No	Reference-deployment evidence	2b1f92ff4481	546f3547b727

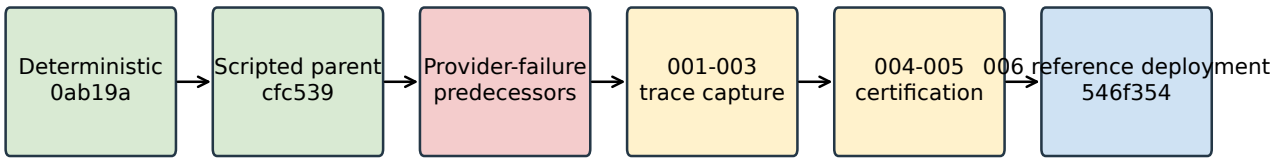


Figure 9: Artifact and amendment lineage. Short prefixes identify roots; the source-verification report records full digests and predecessor links.

Table 16: RA-03 eligibility correction. The amended rule accepts the already frozen benign metadata capability; deterministic causal counts do not use RA-03.

Rule	Attempts	Eligible
Pre-correction global rule	60	0
Amendment 003 permitted-capability rule	20	20

No figure interpolates a causal effect between these measured placements.

The selected utility result is intentionally reported alongside security benefit. Restriction expiry and recovery remain receiver-local, and the artifact retains the raw treatment rows required to inspect the completion calculation.

F Failure-Control Detail

Ten executed live controls passed: invalid signature, wrong receipt, expired context, replayed context, monitor-only policy, gateway cache disabled, receiver unavailability, authorization timeout, stale decision, and arrival after local containment. Unknown peer and remote TCX action were not injected live; executed signed-envelope mutations separately cover unknown-peer rejection and ignored remote action and enforcement fields. Disallowed import type, no remote enforcement, gateway restart, and session termination before arrival were recorded as passed without execution and remain untested design properties. Appendix E preserves all source-control rows and their evidence classifications; Table 7 distinguishes the executed mutation corpus. In every evaluated gateway block, recorded provenance is Domain-B-local; the literal-zero remote-enforcement counter supplies no executed misuse test. The artifact's `docs/KNOWN_ISSUES.md` records the correction without changing the sealed source.

G Reproducibility Details

The default reviewer workflow avoids live credentials. It verifies source roots, regenerates all paper-local derivatives, and

Table 17: What the evaluation supports and does not support.

Evidence	Supported scope
Deterministic matched pairs	Causal comparison under matched frozen local policy and synthetic scenarios
Executable replay	Exact captured action-sequence comparison without model credentials
Live trace generation	A bounded credentialed cohort generated action attempts
Single-environment reference-path deployment	A signed tcopd-A-to-tcopd-B path and receiver-local gateway decisions in one reference environment
Validation-value and assurance cohort	Semantic binding and executed receiver-validator mutations; hostile-peer and correlation tables are formula-computed illustrations
Post-denial substitution replay	Policy-conditioned post-denial branches in a bounded reference runtime, not deployment prevalence
Campaign Correlation frontier	Deterministic receiver-local Campaign Correlation selectivity, not a general escalation policy or deployment-effectiveness estimate
Full external-warning ledger	All 168 sealed external source cases: 25 accepted exact bindings, 47 attack-bearing no-warning cases, and 96 benign no-warning cases; authorization coverage, not detector-quality evidence
Not established	Production deployment, Internet-scale federation, key-lifecycle operation, uninstrumented incidents, or universal superiority to central defense

Table 18: Strict replay timing treatments. A treatment can be too late for one action while still affecting later opportunities.

Action-relative placement	Improved	Unchanged	Harmful-action delta
Before first harmful opportunity	24	20	-72
At first-action authorization boundary	24	20	-72
After initial harm, before later actions	24	20	-48
After receiver-local detection	24	20	-48
After partial local containment	12	32	-24

invokes strict replay on frozen traces. Exact expected counts and full source digests are machine-checkable in the anonymous artifact manifest. The deterministic causal-core tier writes a new output root and compares its declared deterministic results; it never overwrites the sealed sources.

The live provider environment is documented only for optional trace regeneration. A current provider response can differ from the frozen cohort; such a replication does not replace the trace-replay evidence or modify the paper's claims.

Table 19: Negative and failure controls. Ten controls (NC-01–04, NC-07, NC-10–12, NC-14, NC-16) were executed and passed in the sealed live artifact. NC-05 and NC-09 are covered by executed signed-envelope mutations in `artifacts/tcx-validation-value-v2/property-test-report.json` (`issuer` and `action_semantic`, respectively), not by live injections. Four rows (NC-06, NC-08, NC-13, NC-15) were recorded as passed in the sealed live artifact without executing the condition; they are untested design properties, not results. See the artifact's known-issues file, `docs/KNOWN_ISSUES.md`.

Condition	Expected behavior	Observed behavior	Invariant	Pass	Source / evidence
Invalid signature	Reject before restriction	Rejected; no restriction	Signed input	Yes	NC-01; live executed
Wrong receipt	Reject before restriction	Rejected; no restriction	Receipt correlation	Yes	NC-02; live executed
Expired context	Reject before restriction	Rejected; no restriction	Bounded freshness	Yes	NC-03; live executed
Replayed context	Reject before restriction	Rejected; no restriction	Replay protection	Yes	NC-04; live executed
Unknown peer	Reject before restriction	identity_unknown; no restriction	Authorized peer	Yes	NC-05; mutation: issuer
Disallowed import type	Reject before restriction	Not executed; design property	Context class	Not tested	NC-06; not executed
Monitor-only policy	Do not preempt first call	First call not blocked	Local detection timing	Yes	NC-07; live executed
No remote enforcement	Never accept remote enforcement	Not executed; design property	Receiver authority	Not tested	NC-08; not executed
Remote TCX action	Never interpret remote action	Remote fields action and enforcement ignored	Receiver authority	Yes	NC-09; mutation: action_semantic
Gateway cache disabled	Evaluate each authorization	Cache disabled	Replay correctness	Yes	NC-10; live executed
tcopd B unavailable	Use B-local disposition	Local fail-closed policy	Receiver authority	Yes	NC-11; live executed
Authorization timeout	Use B-local disposition	Sender does not choose timeout	Receiver authority	Yes	NC-12; live executed
Gateway restart	Recompute local state	Not executed; design property	Receiver authority	Not tested	NC-13; not executed
Stale decision	Recompute local state	Fresh decision used	Receiver authority	Yes	NC-14; live executed
Session termination before arrival	Do not cross sessions	Not executed; design property	Session binding	Not tested	NC-15; not executed
Arrival after local containment	No reversal of completed action	Non-preventive late context	Containment window	Yes	NC-16; live executed

Table 20: Local harness performance in microseconds, with 20 samples per selection and component. Authorization uses `perf_counter_ns()` around a Python dictionary cache lookup on hits, or request construction and `evaluator.authorize()` on misses; it does not time an RPC. The enabled selection is labeled fixed 5 s TTL in the artifact, but the timed loop does not exercise expiry. Gateway timing wraps adapter construction and invocation; warning-to-enforcement sums context acceptance and gateway durations. With 20 samples, p95 and p99 are near the top order statistics. These local harness measurements are not production inter-domain latency estimates.

Selection	Samples	Authorization p95 (μs)	Gateway p95 (μs)	Warning-to-enforcement p95 (μs)	Hit rate
Disabled	20	19.75	24.584	331.624	0.0
Enabled, fixed 5 s TTL	20	0.667	27.25	328.208	0.95